HR 2.42, CI 2.07-2.83, p<0.001; Stage II HR 1.27, CI 1.08-1.50, p= 0.004) were associated with worse OS, however, facility type was not.

Conclusion: Better OS was observed at AC compared to CC among patients diagnosed in earlier years and among those with Stage II disease; however, it appears such gap has closed in more recent years and after adjusting for sociodemographic characteristics.

Author Disclosure: **R. Marchant**: None. **X. Hu**: Grant/research funding; St. Jude Children's Research Hospital, Pfizer. **K.A. Ward**: None. **N. Ridge**: None. **N. Ali**: None. **A. McCook-Veal**: None. **J. Switchenko**: None. **P. Santos**: Grant/research funding; Conquer Cancer - The ASCO Foundation. Travel expenses; Conquer Cancer - The ASCO Foundation. **J.S. Remick**: Assist with radiation oncology medical residency program; Emory University.

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Pencil-Beam Proton Therapy for Large-Size and Unfavorably Located Uveal Melanoma: Safety, Toxicity, and Tumor Response

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Purpose/Objective(s): Particle beam therapy (PBS) is a cornerstone treatment for large-size and unfavorably located uveal melanoma (UM), offering ocular and visual preservation as an alternative to enucleation. While passive scattering proton therapy historically served as the gold standard, PBS eliminates the need for patient-specific collimators/compensators, enabling rapid workflows critical for aggressive tumors. As PBS replaces passive scattering globally, its safety and efficacy in high-risk UM require rigorous evaluation.

Materials/Methods: Thirty-eight patients with large-size (median volume: 1.6 cc, range: 0.2−3.44 cc) and unfavorably located (proximity to optic nerve/disc) UM received PBS proton therapy (49 Gy RBE in 7 fractions). Eye immobilization utilized a modified thermoplastic head-mask with gaze fixation on a marker located on focal distance (25 cm). Treatment planning incorporated robust multifield optimization (2 mm setup, 3.5% range uncertainty), 2−3 fields (one with a range-shifter), and a 1 mm margin for eye motion. Movement control was achieved by CBCT and kV-control with every field accompanied by voice commands. Follow-up occurred at a dedicated ophthalmo-oncology center.

Results: Linear and angular median offsets during all treatments from 252 measurements was 0.15 mm (lng) and 0.7°(rtn) respectively, which did not exceed a safe margin. At a median follow-up of 12.5 months (3−54), no local recurrences occurred. Tumor response started with reduced blood flow by ultrasound and structural changes by MRI, with shrinkage observed ≥6 months post-treatment. Early toxicity (dermatitis, keratitis) was ≤Grade 2 by CTCAE v6.0. Late ocular hypertension (OH) due to neovascular glaucoma (NVG) occurred in 12 patients (6−10 months post-treatment). NVG correlated strongly with pre-treatment Bruch's membrane rupture (p=0.0056), macular high dose (p=0.0137), and total visual field loss (p=0.00016). Seven patients (18.4%) required enucleation (median 17 months post-treatment; 6 with uncorrected OH due to NVG).

Conclusion: PBS proton therapy demonstrates promising local control and manageable acute toxicity for high-risk UM. However, NVG-driven ocular hypertension remains a critical late risk, underscoring the importance of macula dose optimization and patient selection—particularly in cases with pre-existing Bruch's membrane rupture or visual field loss. Longer follow-up is needed to validate predictive models for NVG and refine vision-preservation strategies.

Author Disclosure: **N. Martynova**: None. **V. Nikolay**: None. **I.E. Panova**: None. **E.V. Samkovich**: None. **E.M. Svistunova**: None. **G. Andreev**: None. **A. Lyubinskiy**: None. **M. Linnik**: None. **N. Kataev**: None. **E.V. Boyko**: None. **K. Suprun**: None.

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Induction Chemotherapy Using Paclitaxel and Carboplatin followed by Altered Fractionated Radiation Therapy in Locally Advanced Head and Neck Carcinoma

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Purpose/Objective(s): The standard of care treatment for locally advanced head and neck squamous cell carcinoma (HNSCC) is definitive concomitant chemoradiation (CRT). Since, most of these patients have advanced stage disease and compromised performance status, aggressive treatment with CRT is usually not well-tolerated. Induction chemotherapy (IC) may provide a tool to downstage the disease with resultant improved performance status and may provide a good chance for organ preservation. Furthermore, IC bridges the time gap till the start of definitive chemoradiation, as there is long waiting time for radiation therapy appointments. The objective of this study was to assess the disease-free survival (DFS) and overall survival (OS) of IC using paclitaxel and carboplatin followed by altered fractionated radiation therapy (AFRT) in locally advanced HNC.

Materials/Methods: A total of 117 patients, with biopsy proven locally advanced HNSCC were included. Patients with primary tumor in oral cavity (n=30), oropharynx (n=23), larynx (n=29) and hypopharynx (n=35) having tumor, node, metastatic (TNM) stage between III to IVB were enrolled for the study. The patients were scheduled to receive three cycles of IC using paclitaxel and carboplatin every three weeks in a dose of 175 mg/m^2 and AUC=5 respectively on D1 of each cycle. After three cycles of IC, patients were planned for AFRT. A total dose of 55 Gray (Gy) was planned on linear accelerator (at least conformal radiotherapy was delivered) with 2.75 Gy per fraction and five fractions per week. Maximum dose to spinal cord was limited to 33 Gy. The patients, who developed grade 3 or 4 toxicity related to chemotherapy or radiation therapy, were allowed treatment breaks to recover from the respective toxicity. The response was assessed after 6 weeks of completion of treatment using CT scan head and neck with contrast. Primary end point was to assess disease free survival (DFS) while overall survival (OS) was also assessed as a secondary end point.

Results: A total of 109 patients (93%) completed the scheduled treatment and were evaluated while rest of the patients couldn't complete the treatment protocol. Eleven patients (10%) showed disease progression during treatment and were shifted to other therapy. DFS and OS were calculated for a total of 98 patients. Three-year DFS and OS were 42% and 49% respectively. Mean and median DFS (year) were 3.59±0.18 (95% confidence interval [CI] 3.23-3.95) and 3.76±0.32 (CI 3.12-4.40) respectively. Mean and median OS (year) were 4.23±0.15 (CI 3.93-4.53) and 4.32±0.26 (CI 3.80-4.84) respectively.

Conclusion: We concluded that induction chemotherapy using paclitaxel and carboplatin followed by altered fractionated radiation therapy in locally advanced head and neck squamous cell carcinoma is an effective option with reasonable toxicity.

Author Disclosure: **A. Mateen**: None. **M. Mateen**: None. **A. Adil**: None. **F. Abdullah**: None.

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Anxiety, Depression and Fear of Cancer Recurrence (FCR) in HPV-Associated Oropharyngeal Cancer (HPVOPC) Survivors Treated with Chemoradiation (CRT): Implications for Contemporary Clinical Care

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